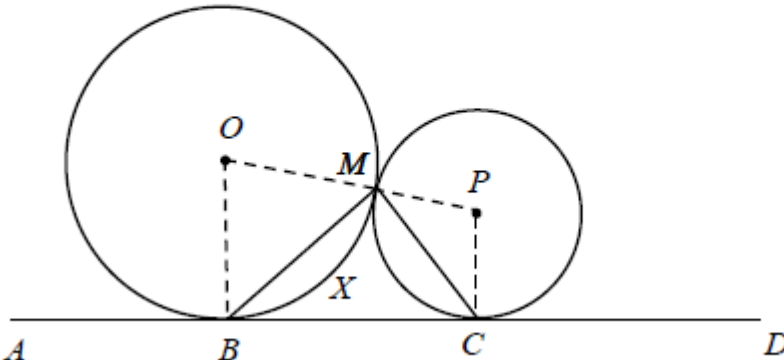
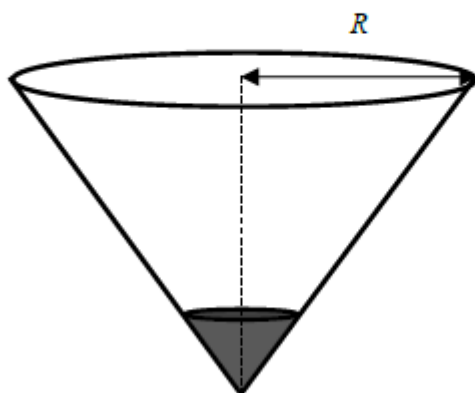


1	YTSS/2025/I/21
	A solid hemisphere with radius r cm was melted to form a solid cylinder with radius $\frac{1}{2}r$ cm. The volume of the hemisphere is 1000 cm^3. (a) Show that r is approximately 7.816 cm. [1] (b) Find the total surface area of the hemisphere. [2] (c) Find the height of the cylinder. [2]
2	YTSS/2025/I/23
	The diagram shows two circles with centres O and P touching each other at M. OP is a straight line and $ABCD$ is the common tangent to the two circles. X is a point on the circumference of the bigger circle.  (a) Show that angle BMC is a right angle. [4] (b) If angle $MBC = 0.6$ radian and chord $BM = 10$ cm, (i) show that acute angle BOM is 1.2 radians, [2] (ii) find the length of arc BXM. [3]
3	YTSS/2025/II/05

The diagram shows a closed inverted cone which contains some water.

The height of the cone is $\frac{4}{3}$ times that of its radius R .

The volume of water is $\frac{1}{125}$ the capacity of the cone.

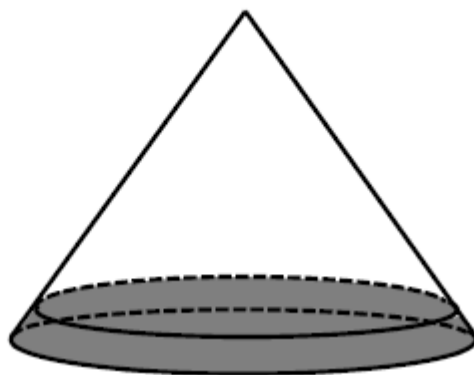


(a) Express the curved surface area of the cone that is covered by the water in terms of R .

[4]

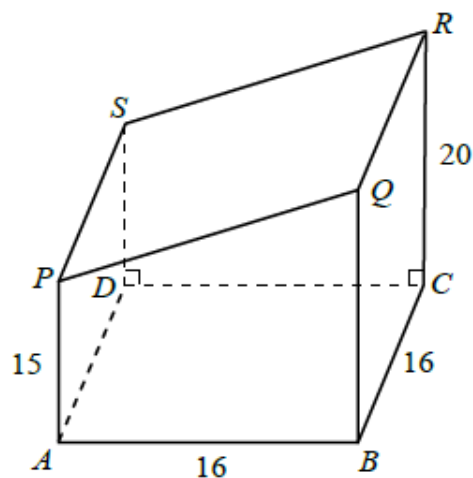
(b) The cone is then inverted as shown in the diagram below.

Given that the height of the cone is 115 cm, find the new depth of the water



[3]

- (a) In the diagram, prism $ABCDSPQR$ is a container.



The base $ABCD$ of the container is a square of side 16 cm.
 $AP = 15$ cm and $CR = 20$ cm.

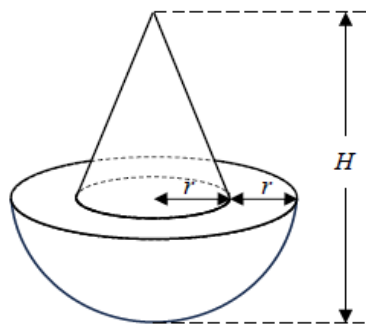
- (i) Calculate the volume of the container.

Answer cm^3 [2]

- (ii) Calculate angle RAC .

[3]

- (b) The diagram shows a solid formed by joining a cone to a hemisphere.



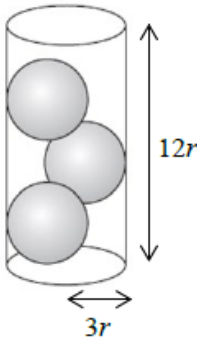
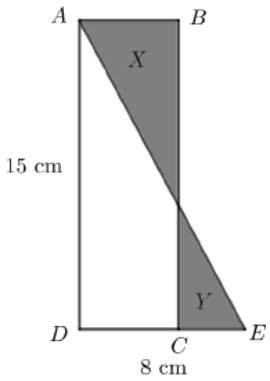
The cone has a base radius of r m.

The hemisphere has a radius of $2r$ m.

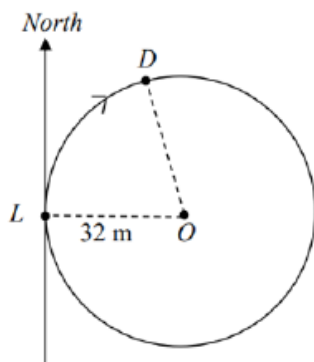
The total height, H , of the solid is 3 times the radius of the hemisphere.

Given that the total volume of the solid is $4.86\pi \text{ m}^3$, calculate the total surface area of the solid.

[3]

5	CCHY/2025/I/16 Three spheres, each of radius $2r$ cm, are placed inside a cylinder. The cylinder has radius $3r$ cm and height $12r$ cm. Water is poured into the cylinder to fill it completely. The volume of water poured into the cylinder is $k\pi r^3 \text{ cm}^3$. Find the value of k.  <i>Answer</i> [3]
6	CCHY/2025/I/20 The figure shows a rectangle $ABCD$ with $AD = 15 \text{ cm}$. E is on DC produced such that $DE = 8 \text{ cm}$. The area of the shaded part X is 12 cm^2 more than the area of the shaded part Y. Find the length of AB.  [3]
7	CCHY/2025/I/22

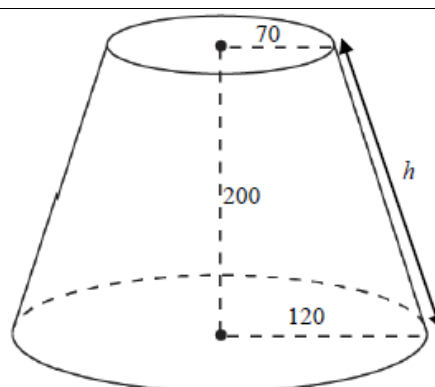
Alex flew an aerial drone from a landing site L . It flew in a circle of radius 32 metres in a clockwise direction shown in the diagram below. The centre of the circle is due east of L .



Find the bearing of the drone D from L after it travelled 43 metres along the circumference of the circle.

[3]

8 SST/2025/I/22



The diagram shows a lampshade in the shape of a frustum.

The bottom radius of the frustum is 120 mm.

The top radius of the frustum is 70 mm.

The vertical height of the frustum is 200 mm.

(a) Find the slant height h of the frustum. [2]

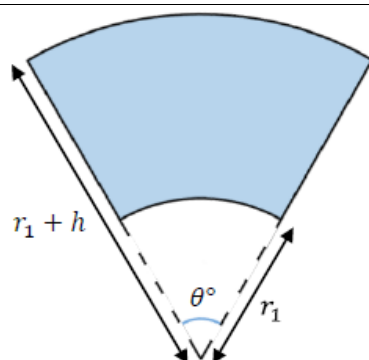
(b) The shaded area represents the net of the lampshade.

The radius of the inner arc is r_1 mm.

The radius of the outer arc is $(r_1 + h)$ mm.

Show that the radius of the inner arc is 288.6 mm.

[3]



(c) Hence, find angle θ . [1]

(d) Calculate the amount of material required to make the lampshade, leaving your answer in square centimeters. [2]

(a)

Figure 1

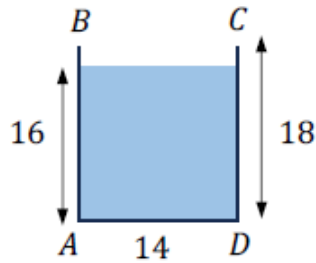


Figure 2

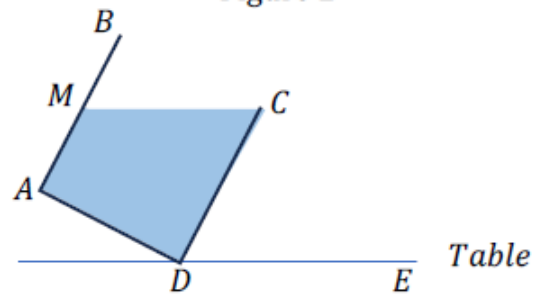


Figure 1 shows one side $ABCD$ of a square-based prism.
Each side of the base is 14 cm. The height of the container DC is 18 cm.
The prism is placed on a table and filled with water until the height of the water reaches 16 cm.
Figure 2 shows the square-based prism tilted until the water level is at MC .
During this process of tilting, some water is spilled out.

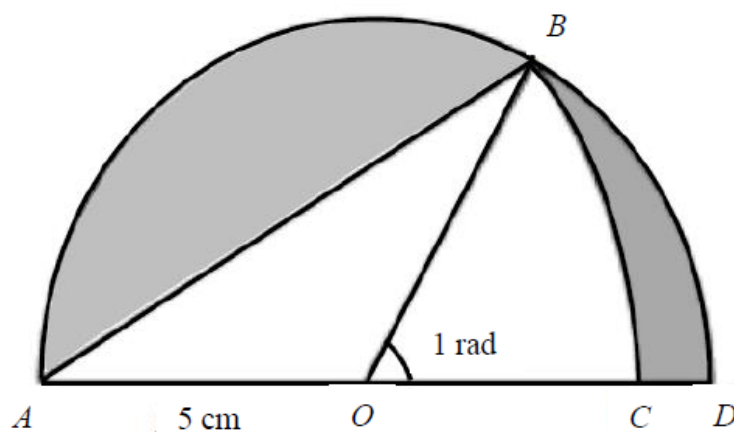
Given that M is the mid-point of AB , find

- (i) the amount of water that was spilled out from the square-based prism during the tilt.

Answer cm^3 [2]

- (ii) angle CDE , the angle between the square-based prism and the table. [2]

$ABDCO$ is a semi-circle with centre O , radius 5 cm and diameter AD .
Arc BC is part of a circle with centre A . Angle $BOC = 1$ radian.



(a) Find the length of arc BD .

Answer cm [1]

(b) Find the length of arc BC .

[3]

(c) Find the perimeter of the shaded region.

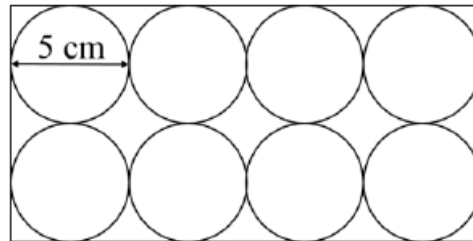
Answer cm [2]

(d) Find the area of the shaded region.

[2]

24 identical cylindrical wax candles, each of diameter 5 cm and height 2 cm, were packed and stacked in a rectangular wooden box.

The diagram shows the top view of the rectangular box containing the candles. The candles touch one another and the sides of the box.



- (a) Find the volume of the rectangular box which was used to pack all the 24 candles.

Answer cm^3 [2]

- (b) Show that the total volume of all the cylindrical candles is $300\pi \text{ cm}^3$.

Answer

[2]

- (c) Find the total surface area of one cylinder candle.

[2]

- (d) All 24 wax candles can be melted and shaped into one spherical candle. Find the total surface area of this spherical candle.

[4]

1	(a) $1000 = \frac{2}{3} \pi r^3 \text{ --- [M1]}$ $r = \sqrt[3]{1000 \div \frac{2}{3} \pi} \text{ --- [M1]}$ $= 7.8159$ $= 7.816(4sf)$ (b) Surface area of hemisphere $= 3\pi(7.816)^2 \text{ --- [M1]}$ $= 576cm^2 \text{ --- [A1]}$ (c) $\pi(0.5 \times 7.816)^2 h = 1000 \text{ --- [M1]}$ $h = 20.8cm(3sf) \text{ --- [A1]}$
2	(a) Let $\angle OBM$ be x $\angle OBC = \angle OCB = 90^\circ$ (tangent $\perp$ radius) --- [M1] $\angle OBM = \angle OMB = x^\circ$ (base angle of isos triangle) $\angle BOM = 180 - 2x$ (Sum of angles in a triangle) --- [M1] $\angle OPC = 360 - 2(90) - (180 - 2x) = 2x$ $\angle PMC$ $= \frac{180 - 2x}{2}$ (sum of angles in a triangle, base angle of isos triangle) $= 90 - x$ $\angle BMC = 180 - x - (90 - x) = 90^\circ$ (adj angles on a st. line) (shown) --- [A1] (bi) $\angle OBM = \frac{\pi}{2} - 0.6 \text{ --- [M1]}$ $\angle BOM = \pi - 2\left(\frac{\pi}{2} - 0.6\right) = 1.2 \text{ --- [A1]}$ (bii) $\sin(0.6) = \frac{5}{OM}$ $OM = \frac{5}{\sin(0.6)}$ $OM = 8.855 \text{ --- [M1]}$ Arc length BXM $= 8.855(1.2) \text{ --- [M1]}$ $= 10.6cm \text{ --- [A1]}$
3	(a)

$$\text{Height of cone} = \frac{4}{3}R$$

$$\begin{aligned}\text{Slant height of cone} &= \sqrt{R^2 + \left(\frac{4}{3}R\right)^2} \\ &= \sqrt{\frac{25}{9}R^2} \quad [\text{M1}] \\ &= \frac{5}{3}R\end{aligned}$$

$$\begin{aligned}\text{Curved surface of cone} &= \pi R \left(\frac{5}{3}R\right) \quad [\text{M1}] \\ &= \frac{5}{3}\pi R^2\end{aligned}$$

$$\begin{aligned}\text{Curved surface of cone covered by water} &= \frac{\frac{5}{3}\pi R^2}{25} \quad [\text{M1}] \\ &= \frac{1}{15}\pi R^2 \\ &\quad [\text{A1}]\end{aligned}$$

(b)

Let height of the empty space be H .

$$V_{\text{empty}} : V_{\text{cone}}$$

$$124 : 125$$

$$H_{\text{empty}} : H_{\text{cone}}$$

$$\sqrt[3]{124} : 5 \quad [\text{M1}] \text{ for ratio of } H_{\text{empty}}$$

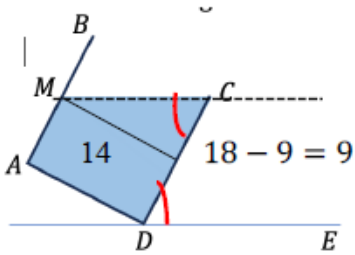
$$5 \text{ units} - 115$$

$$1 \text{ unit} - 23$$

$$\text{Depth of water} = (5 - \sqrt[3]{124}) \times 23 = 0.307 \text{ cm} \quad [\text{M1}][\text{A1}]$$

4	ai aii b	Volume of container $= \frac{1}{2}(15+20)(16) \times 16$ $= 4480 \text{ cm}^3$ $AC = \sqrt{16^2 + 16^2} = \sqrt{512}$ Let the angle be θ. $\tan \theta = \frac{20}{\sqrt{512}}$ $\theta = \tan^{-1}\left(\frac{20}{\sqrt{512}}\right)$ $\approx 41.4729^\circ$ or 0.723839 rad $\therefore \theta = 41.5^\circ$ (1 d.p.) or 0.724 rad (3 s.f.) Height of cone $= 3(2r) - 2r = 4r$ $\frac{1}{3}\pi r^2(4r) + \frac{2}{3}\pi(2r)^3 = 4.86\pi$ $\frac{4}{3}\pi r^3 + \frac{16}{3}\pi r^3 = 4.86\pi$ $\frac{20}{3}\pi r^3 = 4.86\pi$ $r^3 = 0.729$ $r = 0.9$ $\therefore$ Total surface area of solid $= \pi(0.9)\left(\sqrt{0.9^2 + 3.6^2}\right) + \pi(1.8)^2 - \pi(0.9)^2 + 2\pi(1.8)^2$ $= \pi(0.9)\left(\sqrt{13.77}\right) + 2.43\pi + 6.48\pi$ ≈ 38.4836 $= 38.5 \text{ m}^2$ (3 s.f.)
5		Vol of 3 spheres + Vol of water = Vol of cylinder $3 \times \left(\frac{4}{3}\pi(2r)^3\right) + k\pi r^3 = \pi(3r)^3(12r)$ $32\pi r^3 + k\pi r^3 = 108\pi r^3$ $k\pi r^3 = 76\pi r^3$ $\Rightarrow k = 76$
6		Let the unshaded region be Z. Area of X = 12 + Area of Y Area of X + Area of Z = 12 + Area of Y + Area of Z Area of ABCD = Area of $\triangle ADE$ + 12 $AB \times 15 = \frac{1}{2} \times 15 \times 8 + 12$ $AB = \frac{72}{15} = 4.8 \text{ cm}$

7	$\frac{\theta}{360^\circ} \times 2\pi(32) = 43$ $\theta = \frac{43 \times 360^\circ}{2\pi(32)}$ $= 76.991^\circ$ $\angle OLD = \frac{180^\circ - 76.991^\circ}{2}$ $= 51.5045^\circ$ $\text{Bearing} = 090^\circ - 051.5045^\circ$ $= 038.5^\circ$
8	(a) Slant height $= \sqrt{(120 - 70)^2 + 200^2}$ use of Pythagoras' Theorem $= \sqrt{42500}$ $= 206.15$ $= 206 \text{ mm (3 significant figures)}$ (b) $\frac{\theta}{360^\circ} \times 2\pi r_1 = 2\pi(70) \quad \text{---- (1)}$ $\frac{\theta}{360^\circ} \times 2\pi(r_1 + h) = 2\pi(120) \quad \text{---- (2)}$ From (1), $\frac{\theta}{360^\circ} = \frac{2\pi(70)}{2\pi r_1}$ $\frac{\theta}{360^\circ} = \frac{70}{r_1}$ From (2), $\frac{\theta}{360^\circ} = \frac{2\pi(120)}{2\pi(r_1 + h)}$ $\frac{\theta}{360^\circ} = \frac{120}{r_1 + h}$ Hence, $\frac{70}{r_1} = \frac{120}{r_1 + \sqrt{42500}}$ $r_1 = 70\sqrt{17} = 288.6 \text{ mm (shown)}$ (c) $\theta = \frac{2\pi(70) \times 360}{2\pi(70\sqrt{17})} = 87.312^\circ$ $= 87.3^\circ (1 \text{ d.p.})$ (d)

	Total surface area $= \frac{87.312^\circ}{360^\circ} \times \pi(70\sqrt{17} + 50\sqrt{17})^2$ $- \frac{87.312^\circ}{360^\circ} \times \pi(70\sqrt{17})^2$ $= 123053.4615 \text{ mm}^2$ $= 1230 \text{ cm}^2 \text{ (3 significant figures)}$	
9	(ai) Amount of water spilled $= 14^2 \times 16 - \frac{1}{2}(9 + 18) \times 14 \times 14$ $= 3136 - 2646$ $= 490 \text{ cm}^3$ (aia) $\angle MCD = \tan^{-1}\left(\frac{14}{18-9}\right) = 57.264^\circ.$ $\angle CDE = \angle MCD = 57.3^\circ \text{ (1 d.p.)}$ (Alternate angles, <u>MC</u> parallel <u>DE</u>.)  Context: it is about volume NOT area. Context: water level is parallel to the surface of the table.	
10	2a Length of arc $BD = 5(1) = 5 \text{ cm}$	2b $AB^2 = 5^2 + 5^2 - 2(5)(5)\cos(\pi-1)$ $AB = \sqrt{77.015}$ $= 8.7758(5\text{sf})$ Length of arc BC $= 0.5(8.7758)$ $= 4.3879$ $= 4.39 \text{ cm}$ 2c Perimeter $= \text{semicircle} + AB + \text{arc length } BC + CD$ $= 5\pi + 8.7758 + 4.3879 + (10 - 8.7758)$ $= 30.096$ $= 30.1 \text{ cm}^2$ 2d Shaded area $= 0.5\pi(5^2) - 0.5(8.7758^2)(0.5)$ $= 39.270 - 19.254$ $= 20.016$ $= 20.0 \text{ cm}^2$

